

Gliding

Gliding animals are experts at taking off, but do not use muscle power to stay in the air, as flying animals do when they beat their wings. Gliders move through the air on outstretched wings, fins, limbs, or even flaps of skin. Some flying animals also glide—gliding birds can travel immense distances by riding on columns of rising air. Forest animals make shorter glides by jumping off from trees. In the air, they open skin flaps that act as parachutes, speeding them forward as they fall. At sea, flying fish can escape predators by gliding more than 1,000 ft (300 m) on their fins.



Skin flaps create a concave shape that creates lift like a wing

Leap in the dark

Spreading its legs wide, a southern flying squirrel glides between trees on flaps of elastic skin. The folds run from its wrists to its ankles, creating a surface that acts like a wing as it glides through the air. Once it has landed, the flaps tighten up along its sides, so that they do not get in the way as it climbs. Flying squirrels can glide up to 160 ft (50 m) at a time. They live in the forests of North America and, unlike typical squirrels, they feed at night.



Hind legs swing forward, bringing body into vertical position in preparation to land

Fish out of water

After bursting out of the ocean, a flying fish speeds above the surface, gliding on its wing-shaped pectoral fins. Farther down its body, this fish also has a smaller pair of fins that help to keep it in the air. Flying fish have a rapid top speed of about 18 mph (30 kph) and they can glide several yards above the sea. A flying fish's tail works like an outboard motor. If the fish is being chased, it dips its tail into the water and flicks it quickly, accelerating the fish up into another glide.



Giant outstretched foot creates drag, slowing frog's fall

